Project 2

Campbell Smith

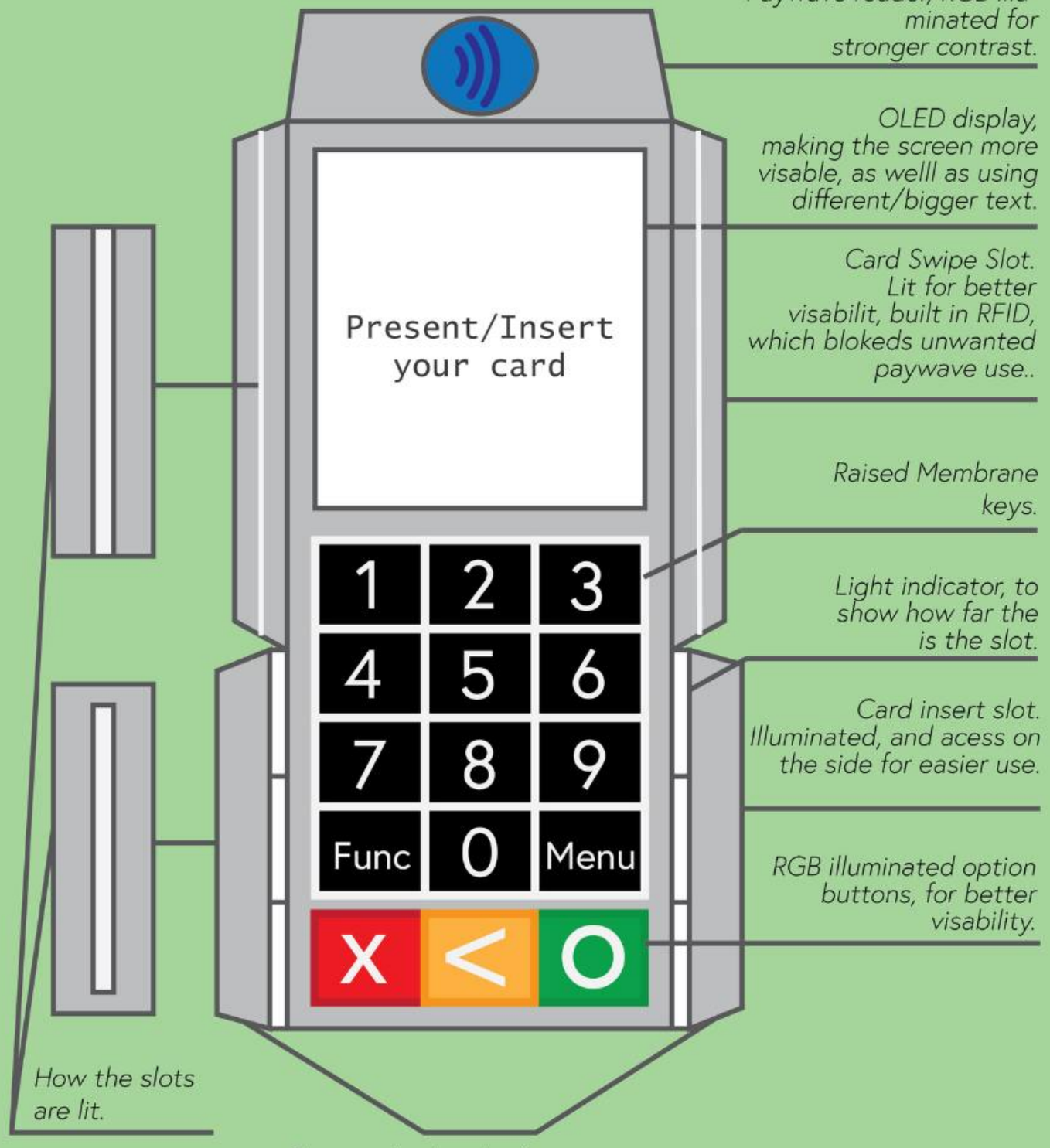
Concept 1, EFTPOS Machine Redesign

In my observation of an elderly gentleman paying for his groceries without 'paywave' I noticed many issues with the EFTPOS machine itself, such as no visuals to where the card slot was, and occasionally difficult to press buttons. For my first concept I wanted to look at how to redesign the EFTPOS machine to suit people with fine motor skill issues, to do this I looked at Fitt's Law.

"The bigger the distance to the target, the longer it will take for the pointer to move to it. The larger the target, the shorter the movement time to it." (Budiu. R. 2022)
I chose to apply Fitt's Law to my design because I found the time in which the EFTPOS interaction took place was a major key pain point, not only for the user, but those around waiting. With time as a driving force it helped me decide what changes would streamline the process of paying with an EFTPOS machine.

My redesign choices were made based on the pain points found in my original research, as well as further observations I made whilst working at a supermarket.
Many of my redesign choices involved making elements of the machine larger, or elements more visible. As I found enlarging elements helped with grip, aiming, and visual communication, and illuminating things such as the card slots, helped as a visual aid of where to direct their card or movement when paying.

Key Details and Overall Design.



Paywave reader, RGB illuminated for stronger contrast.

OLED display, making the screen more visable, as welll as using different/bigger text.

Card Swipe Slot. Lit for better visabilit, built in RFID, which blokeds unwanted paywave use..

Raised Membrane keys.

Light indicator, to show how far the is the slot.

Card insert slot. Illuminated, and acess on the side for easier use.

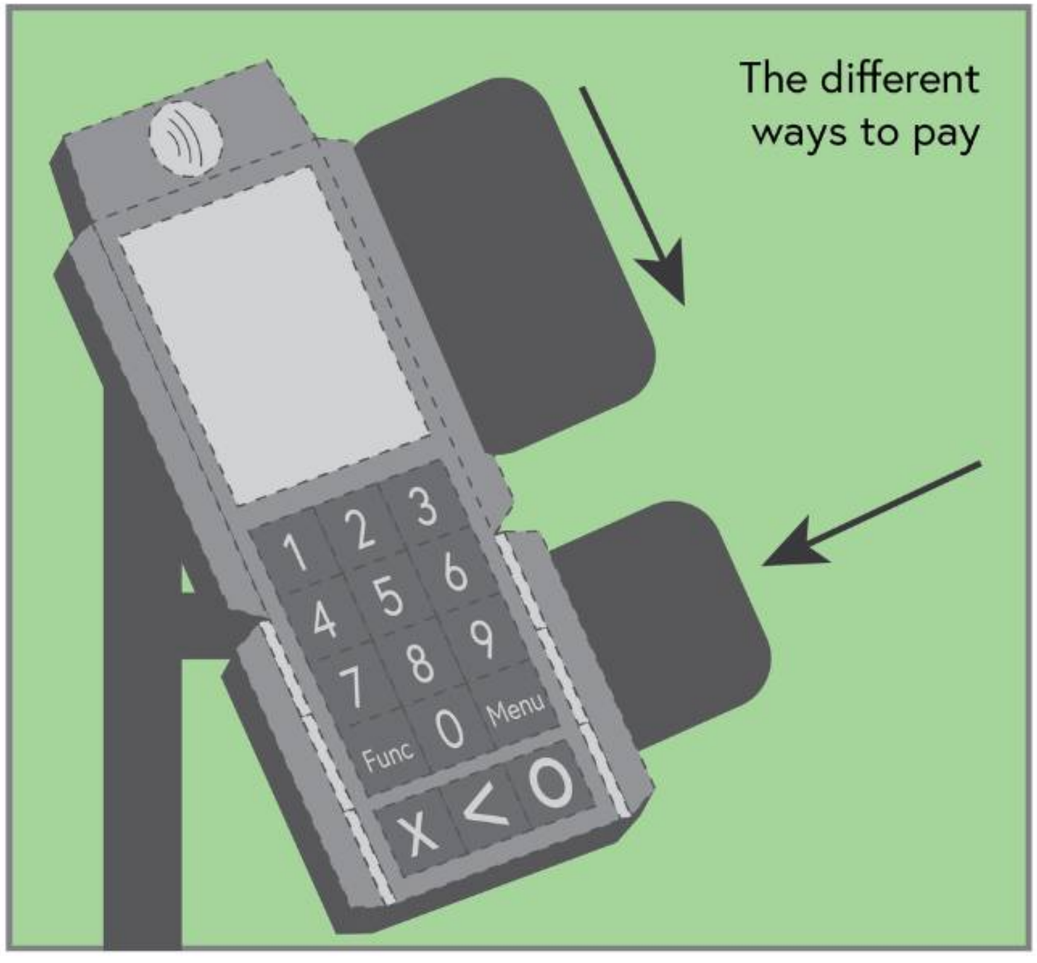
RGB illuminated option buttons, for better visability.

How the slots are lit.

Slots on both sides for easy use, no matter which hand dominate a user is.

For my EFTPOS design I wanted to correct all the issues I saw working at the supermarket, with a primarily older customers. My redesign keeps the core design of the common EFTPOS machine to stop alienation, as well I didn't find many issues with the basic shape.
A majority of the pain points I found were to do with visablity of card slots, and text. By making things more visable with illumination this reduces this issue. As well I introduce slots on both sides of the machine, meaning no matter which hand the user uses they can approach the machine without twisting into uncomfortable positions.
Most estenial from this redesign, is that to make a more inclusive EFTPOS machine that is easier to use for those with motor skill issues, there isn't much a manufacturer would have to change. My resolutions are cost effective, and only require a little more space as the machine is wider, and a little taller.

The different ways to pay



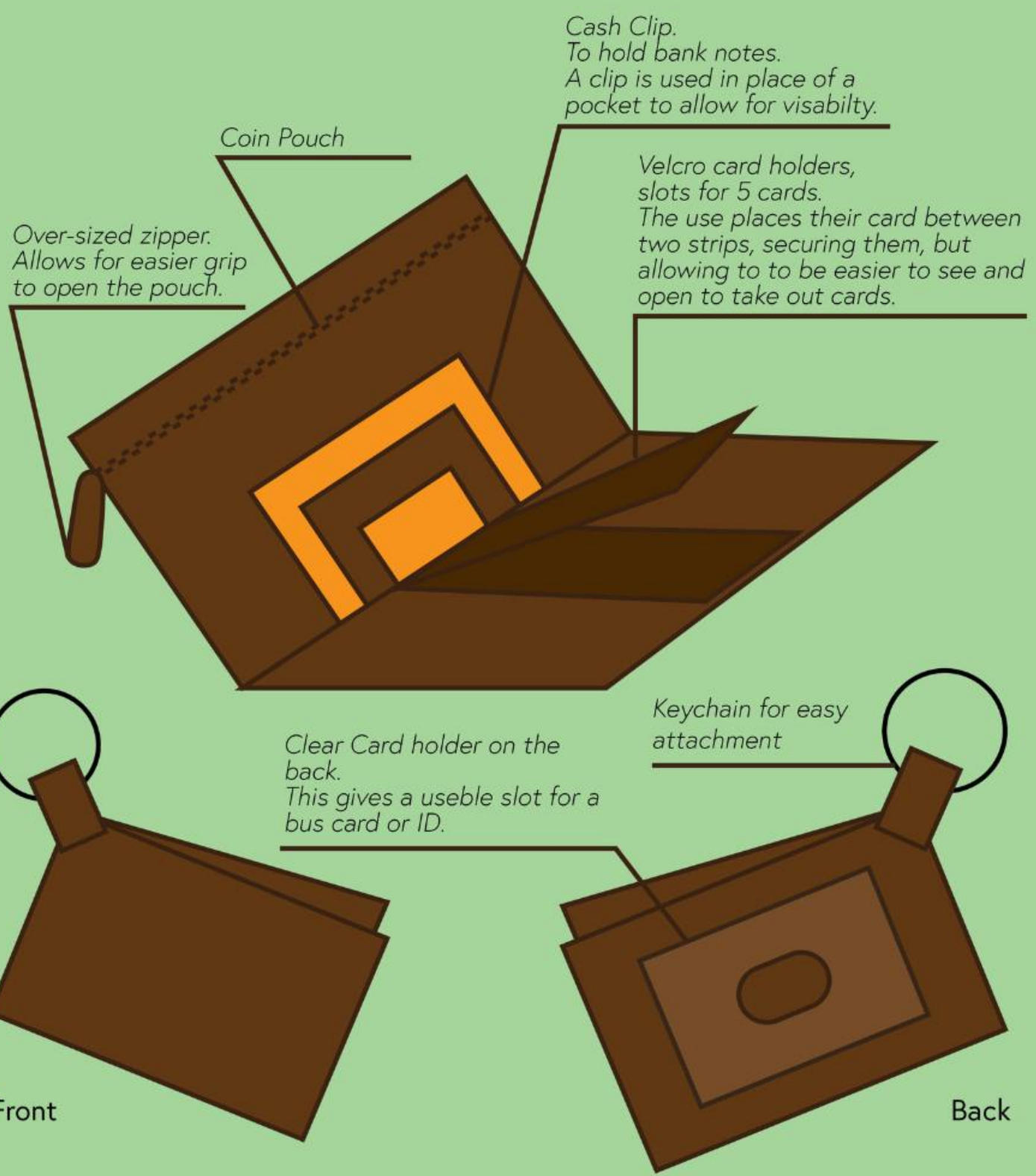
Concept 2, Wallet Redesign

One part of my EFTPOS observation I previously overlooked was the gentlemans struggle with his wallet. The wallet design was the typical man's wallet design, 6 card slots, a coin pouch, and a slit across to hold cash. The card slots however were so stiff the man struggled to get out his card, and couldn't seem to grip the zipper on the coin pouch, so I decided to look at a complete redesign of the wallet for someone with these sorts of fine motor issues. A further point I have noticed, is with as is the loss of memory, so as a part of this design I looked into Miller's Law.

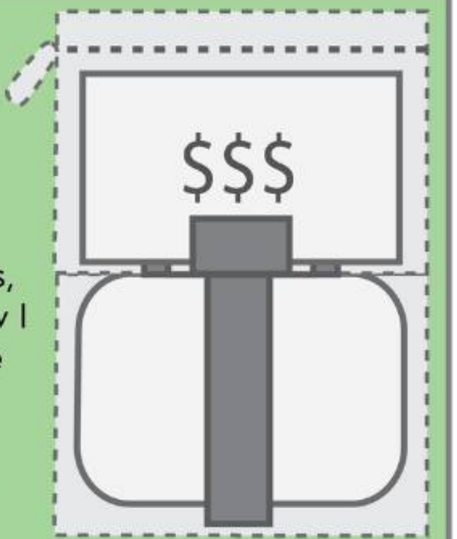
"Miller's Law asserts that the immediate memory span of people is limited to approximately seven items, plus or minus two."(Greenan. R. 2023)I applied Miller's law not only due to the typical age demographic of those with motor issues, but also to help limit myself to keep the design simple and easy to follow, no matter the user. I stuck to 7 visible places for the user to keep things, with 5 card slots, 1 coin pouch, and 1 money clip, on the inside of the wallet.

Key choices I made in my design were around the coin pouch, money clip, and card slots. The first push for my to look into how to redesign the wallet was the man struggle with the card slots, so I looked into a way to create less tension on the card, whilst still keeping it secure and stumbled across velcro. The velcro straps I introduced hold the cards in place, with more visibility to the user, only covering a band in the middle of the card as opposed to the typical wallet which covers a good 80% leaving only the top visible. Not only that, they are easier to pull apart, and the silicone dots on them create friction to stop the cards from sliding. My zipper for the coin pouch is oversized to create more area to grip. And the money clip keeps the banknotes more visible than the typical bank not pocket in a wallet. These things added together create the 7 visible places on the inside of the wallet for the user, with a clear card slot on the outside for those things the user may not always want to open their wallet for.

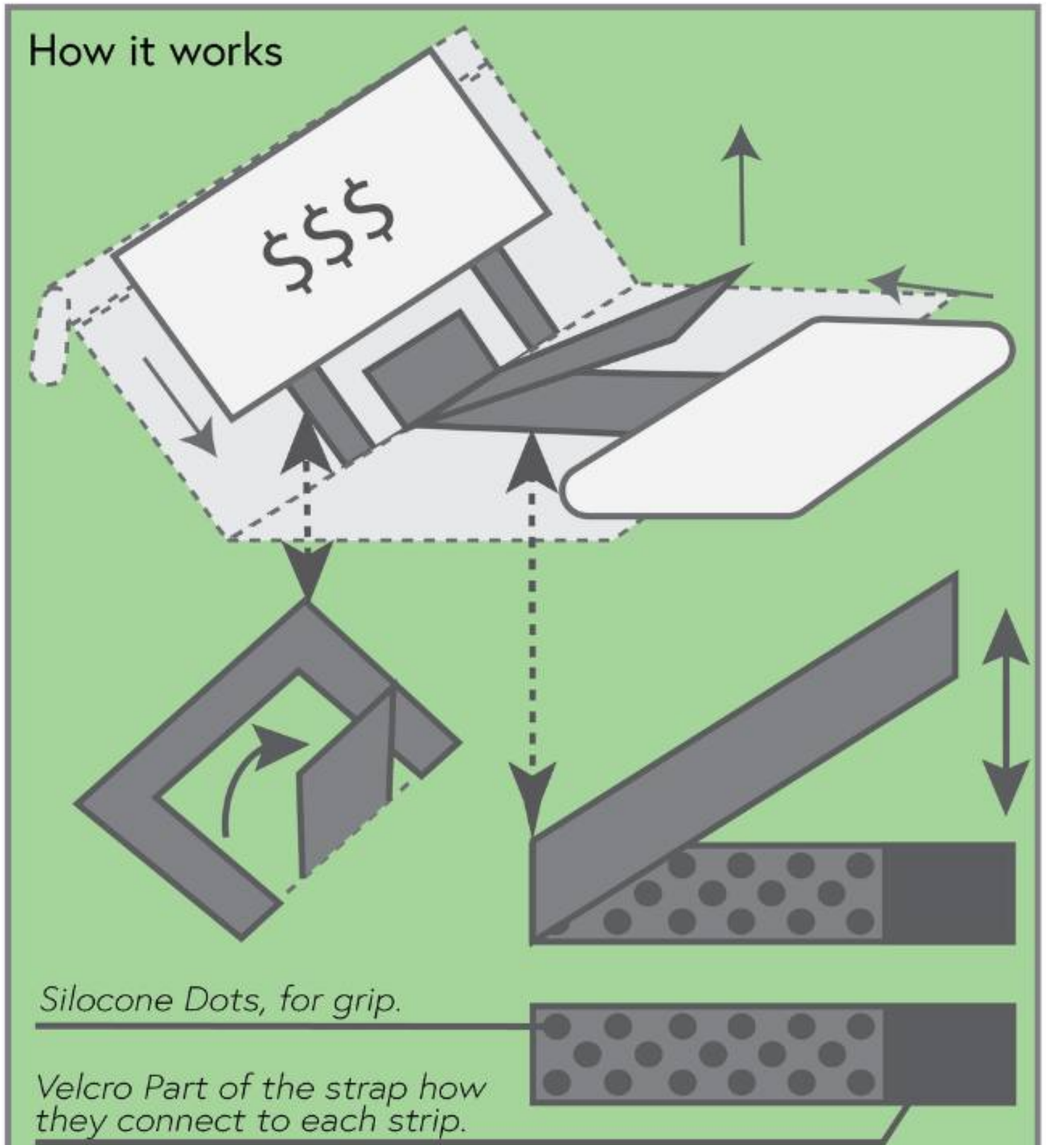
Key Details and Overall Design.



My wallet design was a heavily researched solution to a pain point I didn't really touch on in my EFTPOS observation, but found in further research.
I address these issues seen in the common wallet with the velcro straps, and the money clip. Using MIller's law I created 7 spaces for the user to have items, a money clip for cash, a coin purse for loose change, and 5 strap based card slots which are easier to get cards from than normal slots.



How it works



Concept 3, Coffee Ordering App

My Coffee ordering app concept came from my second interaction, in which a member of the hall wanted to get a hot chocolate, but came across many different steps in which it took for them to get their drink, such as walking outside, bringing their own mug, and having to bring their own supplies. For this concept I wanted to design an app to cut out some of the discomfort, and more tedious parts of making a warm drink for residents. I choose to use Aesthetic Usability for this design.

"The aesthetic usability effect refers to users' tendency to perceive attractive products as more usable. People tend to believe that things that look better will work better — even if they aren't actually more effective or efficient." (Moran. K. 2017)I wanted to look at Aesthetic Usability to help with streamlining the look, making it visually appealing to the user so it takes out some of the stress they may have over the many choices they could make, especially when in menus that may have them look through many options. Aesthetic usability, simplifies things, and helps to not over stimulate the user.

My app focuses on a way for the residents to order their warm drinks to their liking, for delivery or pick-up from their own room, also offering a mug collection service so residents don't have to take the 2 minute walk back to the kitchen. I put a lot of emphasis into the choices the user could make, as an issue I saw in the original observation was the resident having to bring everything they wanted with them, such as additional cocoa powder. The app offers a streamlined service for residents that allows them to relax, and not stress over something like timing the perfect no queue for the coffee machine, instead they can order their drink and pick it up when their ready to leave their warm bed.

Main Page

Pick!

Good Morning, XXXX!

In Need of a Pick me up?

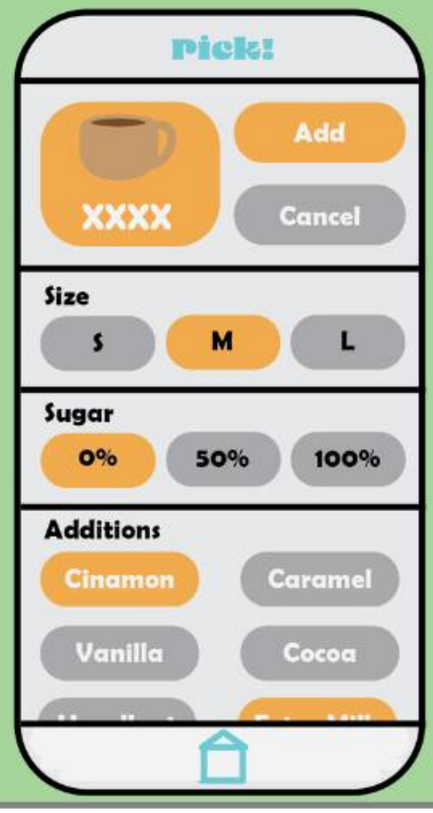
Order

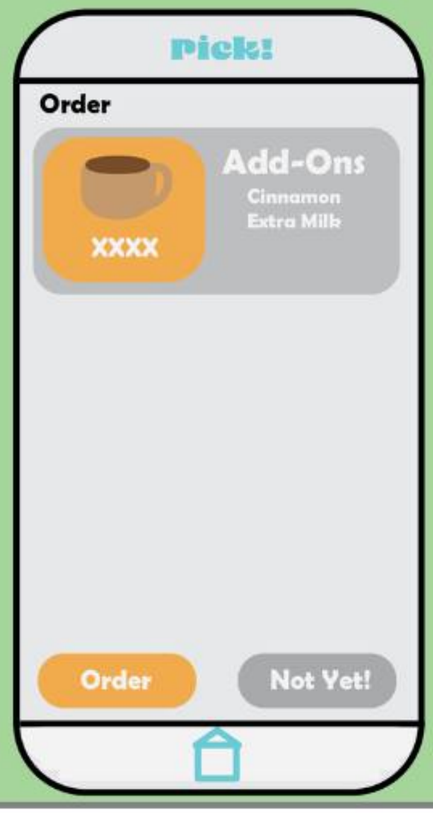
Got a mug?

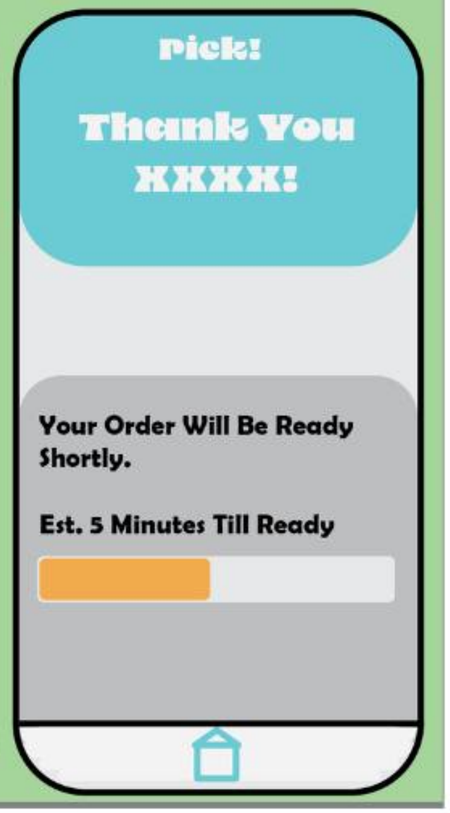
Mug Collection Times

Ordering



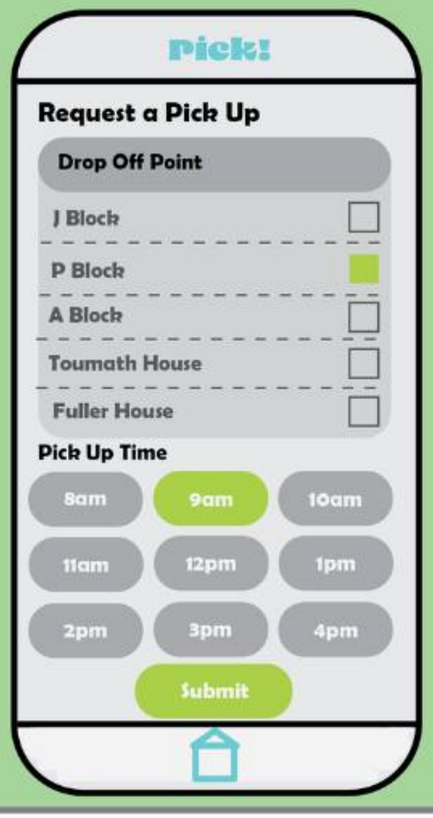


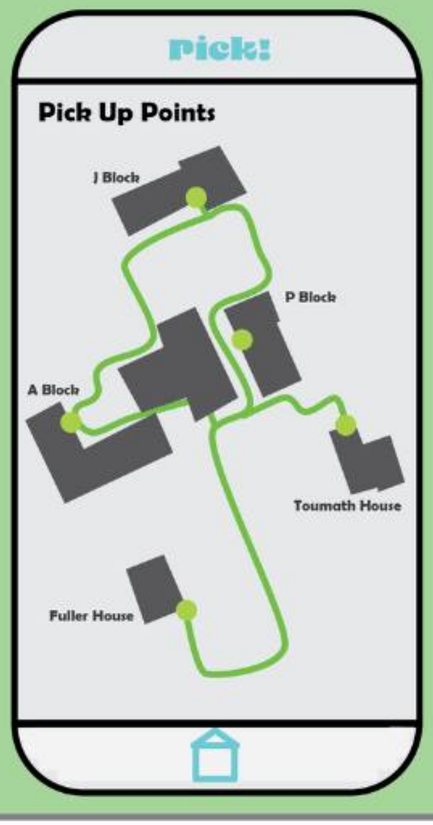




Mug Collection







My app serves as a solution to my pain point around the travel distance between the users room and the kitchen to pick up hot drinks.

My app offers a solution in a way to order the drink as well as a delivery service, that includes a mug pick-up when the user is finished.

The mug pick up has sheduled times in which someone comes to collect the mug bin, but also can be booked if someone were to notice an overflow.

Bibliography

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